EXPT-10

12-09-26

INTERNET WORKING WITH ROUTERS IN CISCO PACKET TRACER

AIM:-

To design and configure a simple internet work using a router and observe network connectivity between two PCs.

STEPS:-

1. Create the network topology
   - ↳ Open cisco packet tracer and create a new project.
   - ↳ Place one router and two PCs.
   - ↳ Connect PC0 and PC1 to the router using copper straight-through cables.

2. Configure the Router:
   - ↳ Select the router and open CLI
   - ↳ Press Enter.
   - ↳ Type enable to enter priviledged mode.
   - ↳ Configure the router interfaces as follow:

```
Router>enable.
Router# config t.
Router(config)# interface FastEthernet 0/0.
Router(config-if)# ip address 192.168.10.1 255.255.255.0.
Router(config-if)# no shutdown.

Router(config)# interface FastEthernet 0/1.
Router(config-if)# ip address 192.168.20.1 255.255.255.0.
Router(config-if)# no shutdown.
```

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between two

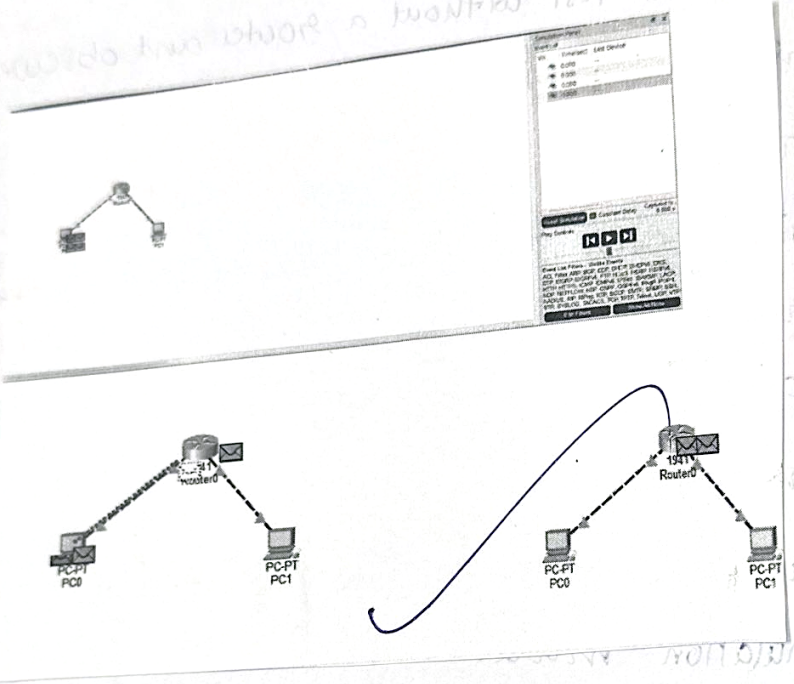
STEPS:-

1. Create the
   - Open cisco
   - Place one
   - Connect
     through

2. Configure
   - Select
   - Press
   - Type
   - Confi

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3. Configure the PCs:-

↳ Open each PC -> Desktop -> IP Configuration.

↳ Assign the following addresses:

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1.

4. Connect PCs with Router.

↳ Connect PC0 -> FastEthernet 0/0 of Router1.

↳ Connect PC1 -> FastEthernet 0/1 of Router1.

↳ Verify that the links become active.

5. Simulate Packet

↳ Select Simulation mode.

↳ Use the simple PDU tool.

↳ Send a packet from PC0 to PC1.

↳ Observe the packet travelling through Router1.

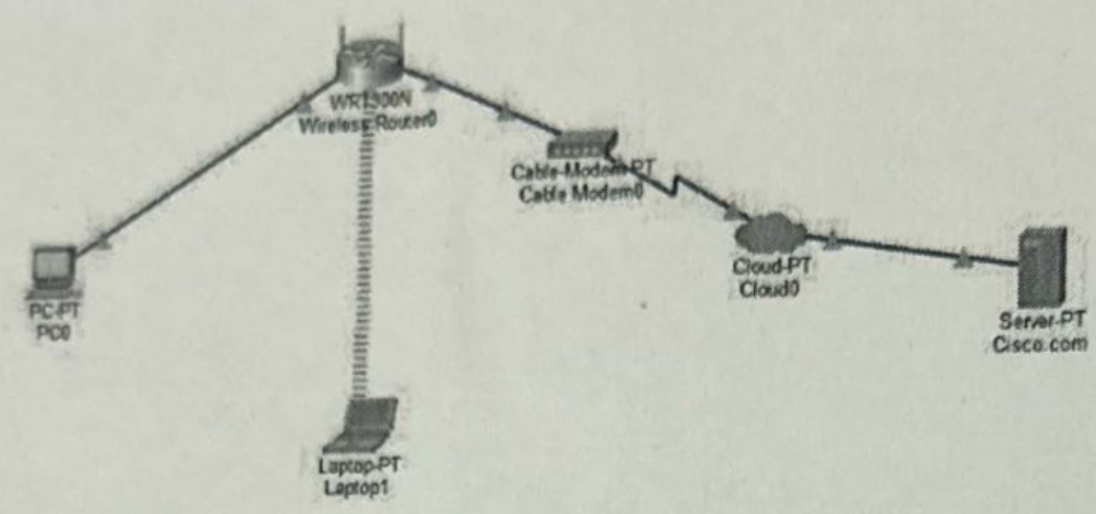
6. Check Connectivity:

↳ Observe the succesful packet transmission between PC0 and PC1.

↳ Verify that the router forwards the packet from the 192.168.10.0 network to the 192.168.20.0 network.

RESULT:-

Thus, a simple internetwork consisting of two PCs and a router was succesfully designed and configured in Cisco Packet Tracer and packet transmission from PC0 and PC1 was succesfully observed.




```
Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.96EE.2B1B
   Link-local IPv6 Address.........: FE80::201:96FF:FEEE:2B1B
   IP Address......................: 192.168.0.101
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.0.1
   DNS Servers.....................: 208.67.220.220
   DHCP Servers....................: 192.168.0.1
   DHCPv6 Client DUID..............: 00-01-00-01-7D-90-28-DC-00-01-96-EE-2B-1B
```

```
C:\>ping Cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=10ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```